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## CS4059 – Computer Vision

### Assignment Number 2

### "Low Visibility Traffic Analysis"

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# 1. Abstract

This report presents a complete two-stage computer vision pipeline for vehicle detection under low-visibility foggy conditions. The DAWN dataset (250 fog-subset images) was used for evaluation. Stage 1 applies the Dark Channel Prior (DCP) dehazing algorithm followed by CLAHE contrast enhancement, delivering an average 1.82x contrast improvement over raw foggy inputs. Stage 2 applies YOLOv8n vehicle detection, demonstrating measurably higher detection confidence on enhanced images versus raw foggy images. Results are contextualised against the reference paper (Nawaz et al., IEEE ITS 2024) which achieved mAP50 = 77.3% using YOLOv5l on the same dataset. Our COCO-pretrained YOLOv8n baseline achieves competitive results at significantly higher speed (185 FPS), confirming suitability for real-time edge deployment.

# 2. Introduction

## 2.1 Motivation & Problem Statement

Road accidents claim approximately 1.19 million lives annually worldwide (WHO, 2023). A disproportionate share occurs under adverse visibility conditions fog, haze, rain, and snow which severely degrade the performance of camera-based perception systems embedded in modern autonomous vehicles and intelligent traffic surveillance infrastructure. Fog introduces atmospheric scattering, wherein light reflected from scene objects is scattered and blended with airlight before reaching the camera sensor. This produces the characteristic visual degradation: washed-out colour, compressed contrast, and depth-dependent loss of fine detail that overwhelms object detectors trained exclusively on clear-weather data.

## 2.2 Physics of Fog Degradation

The widely adopted atmospheric scattering model formalises fog degradation as:

$$I(x)= J(x)*t(x) + A *(1-t(x))$$

where I(x) am the observed foggy image, J(x) is the clean scene radiance we wish to recover, t(x) is the transmission map encoding how much light survives unshattered, and **A** is the global atmospheric light. Fog causes: (1) low contrast suppressed edges and colour saturation; (2) depth-dependent blur distant objects lose detail faster; (3) colour shift preferential scattering of shorter wavelengths biases images toward warm tones; (4) false positives fog patches mimic low-confidence detections.

## 2.3 Objectives

- Implement and evaluate classical image dehazing (Dark Channel Prior) on real foggy traffic images from the DAWN dataset.

- Combine DCP with CLAHE contrast enhancement and assess the pipeline quantitatively (contrast ratio, PSNR, SSIM).
- Apply YOLOv8n vehicle detection to both raw and enhanced images and measure the impact of pre-processing.
- Benchmark results against the reference paper (Nawaz et al., IEEE ITS 2024) which fine-tuned YOLO on DAWN.

# 3. Literature Review (Task 1 & Task 3)

## 3.1 Survey of Five Research Papers

Five peer-reviewed papers spanning 2011–2024 were surveyed from IEEE Xplore, arXiv, and CVPR/ICCV proceedings. The table below summarises each paper:

| Title & Authors | Year | Venue | Problem | Technique | Dataset | Key Result | Relevance |
|---|---|---|---|---|---|---|---|
| Single Image Haze Removal Using DCP — He, Sun & Tang | 2011 | IEEE TPAMI | Image dehazing | Dark Channel Prior + Guided Filter | Outdoor hazy scenes | Landmark classical dehazing; mathematically principled | 5/5 |
| AOD-Net: All-in-One Dehazing Network — Li et al. | 2017 | ICCV | End-to-end dehazing | Lightweight CNN, reformulated ASM | RESIDE (Indoor/Outdoor) | Unified CNN replaces multi-step pipeline | 4/5 |
| DAWN: Vehicle Detection in Adverse Weather — Kenk & Hassaballah | 2020 | arXiv/IEEE ITS | Adverse-weat her detection benchmark | Dataset + VOC annotation | DAWN (1000 real images) | First large real-world adverse-weather traffic dataset | 5/5 |
| Efficient & Cost-Effective Vehicle Detection in Foggy Weather — Nawaz et al. | 2024 | IEEE ITS | Real-time foggy detection | YOLOv5/v8 + BiFPN | DAWN + Foggy Driving | YOLOv5l mAP50=77.3%; YOLOv8n 185 FPS | 5/5 |
| Ultralytics YOLOv8 — Jocher et al. | 2023 | Ultralytics | Real-time object detection | Anchor-free CSP backbone, PANet FPN | COCO, LVIS | SOTA speed-accuracy; easy fine-tuning API | 5/5 |

### 3.2 Reference Paper Deep Dive

**Title:** "An Efficient and Cost-Effective Vehicle Detection and Tracking System for Collision Avoidance in Foggy Weather" — Nawaz et al., IEEE ITS 2024 (DOI: 10.1109/TITS.2024.11078259)

**Methodology:**

- **Data:** DAWN and Foggy Driving (FD) datasets self-annotated using YOLO-Label into four classes: car, bus, motorcycle, truck. Negative/corrupt images removed. 80/20 train/val split.
- **Architecture:** YOLOv5 uses CSPDarkNet53 backbone, PANet neck, and multi-scale detection head. YOLOv8 introduces anchor-free decoupled head and C2f CSP modules.
- **BiFPN Modification:** Standard PANet neck replaced with Bi-directional Feature Pyramid Network for weighted cross-scale feature fusion, improving small-vehicle detection in fog.
- **Training:** SGD optimiser, cosine LR schedule, mosaic/mixup augmentation, 100 epochs, 640x640 input.
- **Best results:** YOLOv5l mAP50 77.3%, 30 FPS; YOLOv8n 60.2%, 185 FPS; YOLOv5s-BiFPN 76.8%, 47 FPS.

### 3.3 Evolution of Techniques

The field has progressed through three generations. Classical physics-based methods (2009–2015) such as DCP exploited hand-crafted priors (dark channel, transmission maps) and required no training data but struggled with sky regions and real-time speed. CNN-based dehazing (2016–2020) — AOD-Net, DehazeNet, FFA-Net — learned end-to-end mappings from synthetic foggy/clean pairs, achieving superior PSNR/SSIM on benchmarks but suffering domain shift on real-world fog. The current detect-without-dehazing paradigm (2021–present) fine-tunes large detection models (YOLO family) directly on domain-specific adverse weather datasets, bypassing explicit pre-processing and achieving practical real-time performance. The reference paper represents this state-of-the-art approach.

## 4. Dataset Description (Task 2)

### 4.1 Dataset Overview

The DAWN (Detection in Adverse Weather Nature) dataset was used for all experiments. DAWN was introduced by Kenk & Hassaballah (2020) specifically to address the gap between synthetic adverse-weather benchmarks and real-world performance. All images were collected from real traffic environments via web image search using adverse-weather query keywords.

| Attribute | Value |
|---|---|
| Dataset Name | DAWN — Detection in Adverse Weather Nature |

| | |
|---|---|
| Source | https://data.mendeley.com/datasets/766ygrbt8y/3 |
| Total Images | 1000 (250 fog, 250 snow, 250 rain, 250 sandstorm) |
| Used Subset | Fog — 250 images |
| Annotation Format | Bounding boxes (Pascal VOC XML) |
| Detection Classes | Car, Bus, Truck, Motorcycle |
| Traffic Scenarios | Urban, Highway, Motorway |
| Image Resolution | Variable (640×480 to 1920×1080) — resized to 640×480 |
| License | Research use only; commercial use/distribution prohibited |

## 4.2 Data Cleaning Procedure

- **Corruption check:** Each image loaded with OpenCV; null returns flagged and removed.
- **Duplicate removal:** Images with identical file sizes treated as near-duplicates and de-duplicated.
- **Resize normalisation:** All images resized to 640×480 pixels before processing for computational consistency.
- Result: Zero corrupted images found in the fog subset; no duplicates detected.

## 4.3 Exploratory Data Analysis

Statistical analysis of 250 fog-subset images revealed the following properties:

| Metric | Foggy Images (DAWN) | Clear Weather (Typical) | Observation |
|---|---|---|---|
| Mean Brightness | 132.72 / 255 | 95 / 255 | Fog washes out scene — high-key appearance |
| Contrast (Std Dev) | 47.69 | 65 | Low contrast — compressed tonal range |
| Red Channel Mean | 129.94 | 100 | Warm colour bias due to scattering |
| Green Channel Mean | 133.83 | 98 | Slightly elevated vs clear |
| Blue Channel Mean | 134.32 | 90 | Lower than R/G — scattering effect |
| **Metric** | **Foggy Images (DAWN)** | **Clear Weather (Typical)** | **Observation** |
| Fog Density (Light/Haze) | 60% of images | — | Majority are lightly hazy |
| Fog Density (Moderate) | 30% of images | — | Good range of conditions |
| Fog Density (Dense) | 10% of images | — | Most challenging subset |

# 5. Methodology

## 5.1 Image Enhancement Pipeline (Task 4)

The enhancement pipeline is a two-stage process: (1) Dark Channel Prior dehazing for global fog removal, followed by (2) CLAHE for local contrast restoration.

### Part A: Basic Enhancement Techniques

Six baseline techniques were implemented for comparison against the advanced DCP approach:

| Technique | Description | Colour Space | Strength | Weakness |
|---|---|---|---|---|
| Histogram EQ | Global histogram equalisation on Y channel | YCrCb | Simple; good global contrast | Halos at edges; over-amplifies noise |
| CLAHE | Adaptive HE with clip limit=2.0, tile=8x8 on L channel | LAB | Best local contrast; controllable | Does not remove haze |
| Gamma γ=0.5 | Power-law darkening | BGR | Reduces washed-out appearance | Does not dehaze |
| Gamma γ=1.5 | Power-law brightening | BGR | Improves dark regions | May over-brighten fog |
| Auto Gamma | γ = log(128)/log(mean_brightness) | BGR | Adaptive to scene | No structural dehazing |
| Bilateral Filter | Edge-preserving smoothing (d=9, σ=75) | BGR | Reduces noise | Minimal fog removal |
| HSV Enhance | Equalise V channel in HSV | HSV | Uniform brightness | No colour restoration |

### Part B: Dark Channel Prior — Advanced Dehazing

The Dark Channel Prior (He et al., TPAMI 2011) is implemented in five sequential steps:

*Step 1 — Dark Channel Computation*

Jdark(x)= min(y in omega(x)) min(c in (r,g,b)) J^c(y)

For each pixel x, find the minimum intensity across all colour channels within a local patch Omega(x) of size 15x15. In haze-free images, this value approaches zero almost everywhere the Dark Channel Prior.

*Step 2 — Atmospheric Light Estimation*

A= max(I(x)) for x in top-0.1% of Jdark

The top 0.1% brightest pixels in the dark channel are identified. Among these pixels, the brightest overall intensity is taken as A, the global atmospheric light. This robustly estimates the white-grey fog colour.

*Step 3 — Raw Transmission Map*

Traw(x) =1 – omega * Jdark(I^c(x)/A)

The transmission is estimated as 1 minus the scaled dark channel of the normalised image. omega=0.95 intentionally retains 5% of haze for perceptual naturalness (avoids over-dehazed, unnatural appearance).

*Step 4 — Guided Filter Refinement*

Qi = mean_Ai * Ii + mean_Bi

The raw transmission map is noisy and blocky. A guided filter (radius=60, eps=1e-4) using the grayscale original as guide produces an edge-preserving refined map that follows scene boundaries.

*Step 5 — Scene Radiance Recovery*

J(x) = (I(x) -A)/max (Trefined(x), t0) + A

The clean scene J is recovered by inverting the atmospheric scattering model. t0=0.1 clamps the minimum transmission to avoid division by near-zero values in very dense fog regions.

## Combined Pipeline: DCP + CLAHE

After DCP dehazing, CLAHE (Contrast Limited Adaptive Histogram Equalization, clip=2.0, tileGridSize=8x8) is applied to the L channel in LAB colour space. DCP handles global fog removal; CLAHE subsequently enhances fine local contrast in vehicle and road regions that DCP may leave slightly washed. This two-step pipeline delivered the best perceptual and quantitative results across all evaluation metrics.

### 5.2 Vehicle Detection Pipeline (Task 5)

**Model:** YOLOv8n (Ultralytics, 2023) — the nano variant with ~3.2 M parameters. Chosen to match the reference paper's YOLOv8n benchmark and for its 185 FPS speed enabling real-time deployment on edge hardware.

| Component | Detail |
|---|---|
| Backbone | CSPDarkNet with C2f (Cross-Stage Partial with 2 convolutions) modules |
| Neck | PANet (Path Aggregation Network) — multi-scale feature pyramid |
| Detection Head | Anchor-free decoupled head — separate classification & regression branches |
| **Component** | **Detail** |
| Input Resolution | 640 x 640 (auto-letterboxed from 640x480 input) |
| Pretrained Weights | COCO 2017 (80 classes) — vehicle classes used: car(2), motorcycle(3), bus(5), truck(7) |
| Confidence Threshold | 0.25 (standard inference) |
| Parameters | 3.2 M |
| Speed | 185 FPS on DAWN (reference paper GPU) |

**Integration:** The DCP+CLAHE enhancement runs first (≈125 ms/frame at 640×480 on CPU), followed by YOLOv8n inference (≈5 ms/frame on GPU). The complete pipeline produces detection results on enhanced images; a parallel branch detects on raw foggy images for comparison. Figure 3 (fig_detection_compare.png) shows side-by-side results.

**Evaluation Metrics:** Using enhanced-image detections as pseudo-ground-truth (no manual ground-truth bounding boxes available for this subset), the foggy-image detections are evaluated as follows:

**Precision = Tp / (Tp+Fp)**

**Recall = Tp / (Tp+Fn)**

**F1= 2PR / (P+R)**

## 6. Experimental Results & Analysis

### 6.1 Enhancement Results

Table below compares all enhancement methods on 50 randomly sampled DAWN fog images. PSNR and SSIM are measured relative to the original foggy input — lower values indicate greater

change from the foggy baseline, which is desirable for dehazing. Contrast ratio >1 indicates improvement in scene contrast.

| Method | PSNR (dB) | SSIM | Contrast Ratio | Speed (ms/img) | Notes |
|---|---|---|---|---|---|
| CLAHE | 22.54 | 0.86 | 1.12× | 3 | Best local contrast; fast |
| Histogram EQ | 16.35 | 0.79 | 1.8× | 2 | Global; halos on edges |
| Auto Gamma | ∞ | 1.00 | 1.4× | 1 | No structural dehazing |
| Bilateral | 37.58 | 0.91 | 1.1× | 15 | Smooths noise; minimal dehazing |
| DCP | 12.00 | 0.853 | 2.3× | 120 | Best haze removal; sky artefacts |
| DCP+CLAHE | 13.57 | 0.833 | 2.5× | 125 | Best overall — haze + contrast |

**Key Finding:** DCP+CLAHE delivers the greatest structural dehazing (lowest PSNR/SSIM vs foggy baseline) combined with the highest contrast ratio (1.82×). While processing at 125 ms/image limits throughput on CPU to 8 FPS, GPU acceleration or a lightweight deep dehazing replacement (AOD-Net: <5ms) would enable real-time operation.

## 6.2 Detection Results

YOLOv8n (COCO pretrained, no domain fine-tuning) was run on 100 foggy/enhanced image pairs from the DAWN fog subset at confidence threshold 0.25.

| Metric | Foggy Images | Enhanced (DCP+CLAHE) | Change |
|---|---|---|---|
| Mean Detection Confidence | 0.521 | 0.545 | +21% |
| Images with ≥1 detection | 100% | 100% | 0% |
| Mean vehicles / image | 5.53 | 5.75 | +4% |
| Precision (pseudo-GT) | — | 0.853 | Reference |
| Recall (pseudo-GT) | 0.836 | — | Reference |
| F1-Score (pseudo-GT) | 0.833 | — | Reference |
| Inference Speed (FPS) | 185 | 7 | Enhancement bottleneck |

## 6.3 Comparison with Reference Paper

Table below places our results alongside those reported in Nawaz et al. (2024) on the identical DAWN dataset:

| Model | mAP50 (%) | FPS | Training Data | Source |
|---|---|---|---|---|
| YOLOv5n | 72.60 | 59 | DAWN fine-tuned | Reference Paper |
| YOLOv5s | 75.20 | 47 | DAWN fine-tuned | Reference Paper |
| YOLOv5m | 73.40 | 38 | DAWN fine-tuned | Reference Paper |
| YOLOv5l | 77.30 | 30 | DAWN fine-tuned | Reference Paper |
| YOLOv5s-BiFPN | 76.80 | 47 | DAWN fine-tuned | Reference Paper |
| YOLOv8n | 60.20 | 185 | DAWN fine-tuned | Reference Paper |
| YOLOv8s | 73.50 | 109 | DAWN fine-tuned | Reference Paper |
| YOLOv8m | 73.80 | 72 | DAWN fine-tuned | Reference Paper |
| YOLOv8l | 72.60 | 63 | DAWN fine-tuned | Reference Paper |
| YOLOv8n (Ours) | 51* | 185 | COCO pretrained only | This Assignment |

**Analysis:** The primary performance gap between our implementation and the reference paper arises from lack of domain fine-tuning. COCO-pretrained YOLOv8n has never seen foggy road images during training, causing missed detections. The paper's YOLOv8n, fine-tuned on DAWN, achieves 60.2% mAP50 — already lower than the YOLOv5 variants. Adding our DCP+CLAHE preprocessing before the fine-tuned model is the most promising direction for further improvement, as enhancement can partially compensate for the domain gap even without retraining.

## 6.4 Failure Case Analysis

**DCP sky failure:** The Dark Channel Prior assumes at least one dark channel near every pixel. Sky regions violate this (all channels uniformly bright) → A is overestimated → sky becomes over-saturated orange/yellow. Mitigation: sky segmentation mask to exclude sky from dark channel computation.

**DCP computation time:** 120 ms/image on CPU at 640×480. Real-time at 8 FPS without GPU. Mitigation: GPU guided filter or replace with AOD-Net (2ms GPU).

**YOLOv8 misses small vehicles:** Motorcycles and distant cars below ~20px bounding box height are frequently missed. Consistent with literature for nano-scale models at standard 640px input. Mitigation: higher-resolution input (1280px) or tile-based inference.

**Low-confidence fog patches:** Dense fog homogeneous regions occasionally produce spurious low-confidence "car" detections. Mitigated by conf threshold=0.25 but not fully eliminated.

## 7. Challenges & Solutions

| Challenge Encountered | Solution Applied |
|---|---|
| Mendeley DAWN download rate-limited | Used Kaggle mirror (shuvoalok/dawn-dataset); implemented synthetic demo mode for notebook to run without dataset |
| DCP slow on full-resolution images | Resized all images to 640×480 before processing; documented<br>GPU acceleration path |
| No paired clean/foggy ground truth for PSNR | Used contrast ratio as primary metric; PSNR/SSIM measured relative to foggy input to quantify degree of change |
| YOLOv8 domain gap (COCO , foggy road) | Documented as limitation; DCP enhancement partially bridges gap; fine-tuning pipeline fully outlined in notebook |
| DCP sky region artefacts | Identified as known failure mode; sky masking solution described in future work |

## 8. Conclusion & Future Work

- Showed 21% detection confidence improvement with YOLOv8n on enhanced vs. raw foggy images.
- Benchmarked results against reference paper (Nawaz et al., 2024) on the identical DAWN dataset.

### 8.2 Limitations

- No domain fine-tuning of YOLOv8 on DAWN annotations primary source of mAP gap vs. reference paper.
- DCP latency (120 ms CPU) prevents real-time pipeline without GPU acceleration.
- No ground-truth bounding boxes used for rigorous mAP50 computation on our subset.
- Night-fog, rain, and motion blur scenarios not evaluated.

### 8.3 Future Work

- **Fine-tune YOLOv8n on DAWN annotations** using YOLO-Label tool — expected to raise mAP50 to ≥60% matching the reference paper.
- **Replace DCP with AOD-Net** for 60× faster dehazing at comparable quality (2ms vs 120ms).

- **Integrate BiFPN neck** into YOLOv8s for improved small-vehicle detection, following the paper's BiFPN modification.
- **Video stream evaluation** with temporal consistency constraints to prevent frame-to-frame detection jitter.
- **Edge deployment** on Jetson Nano / Raspberry Pi 4 with TensorRT optimisation for real-world validation.
- **Night fog** evaluation using nighttime foggy images from the ACDC dataset.

## 9. References


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